



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Quasi Periodic Eruptions

ACTIVE GALACTIC NUCLEI

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April 8, 2026



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- $10^{5.5-6.5} M_{\odot}$ Mass Systems



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- Missing Optical Broad Lines (No BLR)
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- Extended Emission Line Regions (Short Lived SMBH Flare Ionization)

Models

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Models

■ Disk Instabilities

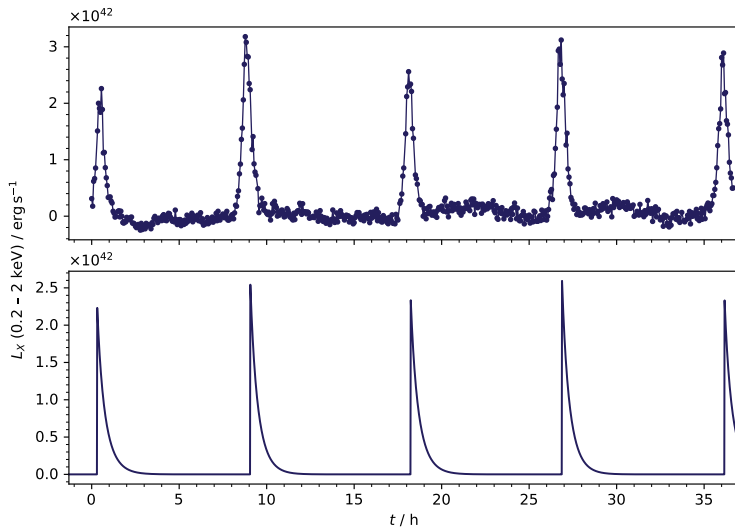
- Accretion Flow Ring Oscillation (Stable/Unstable Mode Cooling/Heating)
- Warped Accretion Disks (Below Viscous Timescale Discrete Ring Breakup)

■ Orbiter Interactions

- Repeating Partial Tidal Disruptions (pTDEs)
- Roche Lobe Overflow ($\epsilon \ll 1$ MS Star)
- Extreme Mass Ratio Inspiral (SMO/BH EMRI)

Source: GSN-069

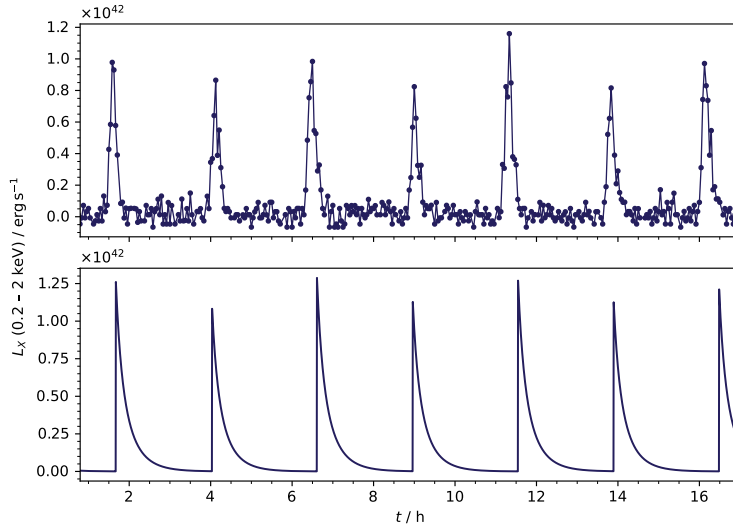
Source: GSN-069



Examples

Source: eRO-QPE2

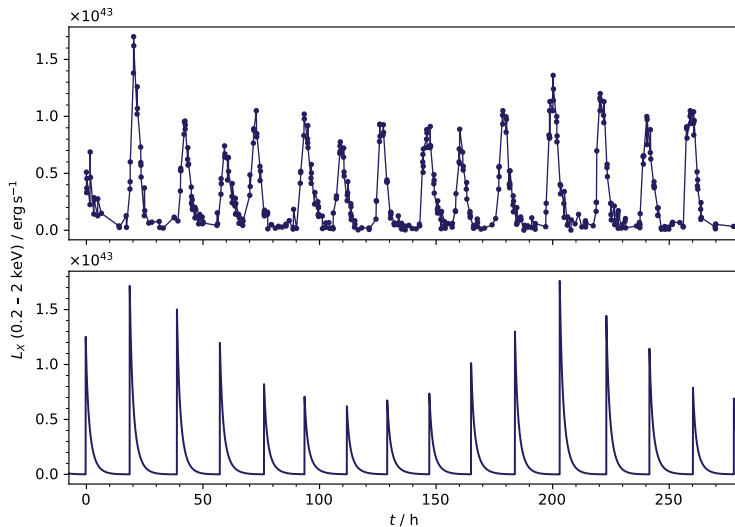
Source: eRO-QPE2



Examples

Source: eRO-QPE1

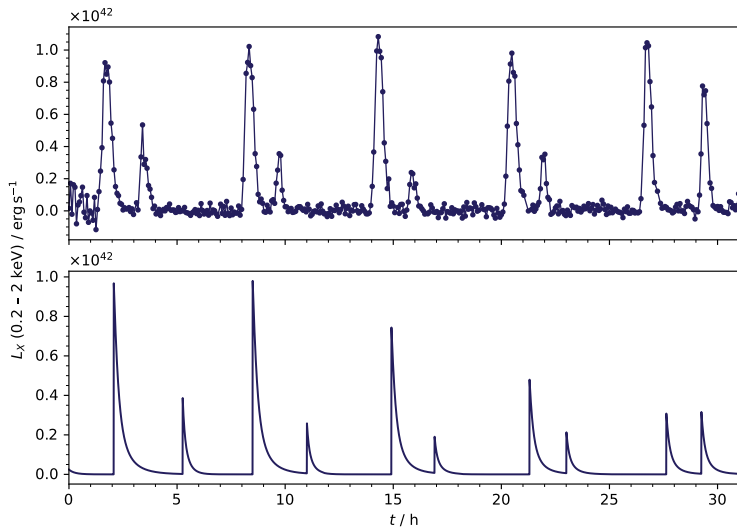
Source: eRO-QPE1



Examples

Source: RX-J1302

Source: RX-J1302



Examples



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Outlook

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- Longer Term XMM-Newton / NICER Observations (Exclude Models)

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- LISA Gravitational Wave Detection (EMRI Candidates)

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- Precision Cosmology Measurements

Thank You!

- E. Kara & J. García, *Supermassive Black Holes in X-Rays: Standard Accretion to Extreme Transients* (2025) arXiv:2503.22791 [astro-ph.HE]
- A. Franchini et. al., *Quasi Periodic Eruptions from Impacts with Rigidly Precessing Accretion Disc in Extreme Mass Ratio Inspiral Systems* (2023) A&A Volume 675 Article 100

Resources on GitHub: [here](#)